UE24CS352A — Machine Learning Mini-Project Report

Sentiment Analysis on Movie Reviews Using TF-IDF & Linear Classifiers

Domain: Natural Language Processing • **Deliverable:** Two-Page Project Summary

1. Problem Statement & Motivation

Sentiment analysis plays a foundational role in modern Natural Language Processing (NLP) by algorithmically decoding human subjective opinions, polarity, and affect from textual communication. In the cinema and entertainment industry, viewer reception heavily impacts box-office trajectories and audience trust. However, unstructured movie reviews present distinct computational challenges: sarcasm, nuanced polarity flips, colloquial syntax, and high lexical sparsity. The core objective of this mini-project is to construct and empirically evaluate a supervised sentiment classification pipeline that accurately categorizes reviews into sentiment classes while benchmarking generative (Multinomial Naive Bayes) versus discriminative linear classifiers (Logistic Regression and Linear Support Vector Machines) using TF-IDF n-gram feature representations.

2. Dataset Characteristics & Preprocessing

The project utilizes the Rotten Tomatoes / Stanford Sentiment Treebank movie review corpus, consisting of **156,059 verified review instances** categorized across **5 distinct classes** (Very Negative, Negative, Neutral, Positive, Very Positive). Pre-encoded or missing instances were checked and handled.

Rigorous Preprocessing Ordering: In adherence to strict machine learning best practices, data partitioning (80% training, 20% holdout test) was executed *prior* to fitting any feature extraction pipeline to eliminate data leakage. The TF-IDF vectorizer extracted unigram and bigram features with sublinear term-frequency scaling (1 + log(TF)), capturing local phrase context (such as 'not good' vs. 'good').

3. Methodology & Classifier Architectures

Three distinct supervised learning paradigms were implemented and benchmarked under identical splits:

- **Multinomial Naive Bayes (MNB):** A generative probabilistic classifier modeling class priors and word-token likelihoods under the conditional feature independence assumption. Laplace smoothing (alpha = 1.0) prevents zero-probability bottlenecks on out-of-vocabulary terms.
- **Logistic Regression (Softmax / L2):** A discriminative linear model minimizing cross-entropy loss with L2 weight regularization. It directly optimizes class posterior probabilities via the logistic sigmoid/softmax function.
- **Linear Support Vector Machine (LinearSVC):** A max-margin discriminative hyperplane classifier minimizing hinge loss with L2 regularization. Platt scaling via 3-fold calibrated cross-validation was incorporated to yield calibrated posterior probability estimates.

(Continued on Page 2: Empirical Results, Comparative Analysis & Demonstration)

4. Experimental Results & Comparative Evaluation

All algorithms were benchmarked using **5-Fold Stratified Cross-Validation** on the training set and evaluated on the unseen 20% holdout test set. Evaluation criteria include Classification Accuracy, Macro F1-score, Weighted F1-score, and Inference Latency per sample.

Model	5-Fold CV Acc	Test Acc	Macro F1	Weighted F1	Latency / Sample
Multinomial Naive Bayes	59.71% ± 0.40%	60.06%	36.49%	54.83%	0.000 ms
Logistic Regression	63.55% ± 0.28%	63.96%	48.24%	61.42%	0.000 ms
Linear SVM	63.09% ± 0.21%	63.85%	50.27%	61.42%	0.008 ms

5. Discussion & Theoretical Analysis

VC Dimension and Overfitting Trade-Off: The empirical findings align closely with theoretical expectations outlined by Vapnik-Chervonenkis (VC) dimension theory. The high-dimensional feature space resulting from unigram + bigram representations makes linear models like Logistic Regression and Linear SVM highly expressive, achieving high boundary separation while retaining strong test generalization via L2 regularization. In contrast, while Naive Bayes is computationally rapid (lowest training duration and inference latency), its strong independence assumption makes it susceptible to feature correlation distortion, slightly lagging discriminative models on overall accuracy.

Champion Model Selection & Metric Prioritization: Linear SVM is selected as the champion model when Macro F1 is the primary selection criterion, achieving **50.27% Macro F1** (compared to 48.24% for Logistic Regression and 36.49% for Naive Bayes). While Logistic Regression achieves the highest test accuracy (**63.96%** vs. 63.85% for Linear SVM), the selection framework intentionally prioritizes Macro F1 over test accuracy alone. Because the dataset exhibits significant class imbalance—with Neutral reviews accounting for over 51% of all instances—raw test accuracy is heavily driven by majority-class predictions. Macro F1 calculates the unweighted average across all five sentiment classes, directly measuring performance across minority polar classes (Very Negative and Very Positive) rather than rewarding majority-class bias.

Inference Latency & Production Viability: All three algorithms achieve sub-millisecond per-sample inference latency, satisfying high-throughput interactive application demands. Linear SVM delivers the optimal production trade-off, combining top class-balanced generalization with calibrated 5-class posterior probabilities.

6. Interactive Terminal Interface & Modular Architecture

To enable real-time demonstration during the technical review, a clean terminal-based interface (main.py) was developed. It features: (1) real-time sentiment inference across all five classes with genuine probability scoring, (2) dynamic TF-IDF token explainability showing top contributing n-grams, (3) comprehensive multi-model benchmark visualization with 5×5 confusion matrices, and (4) instant deliverable exports. The repository is organized into modular Python files adhering to clean coding standards.

7. Conclusions & Future Work

This project successfully demonstrated the effectiveness of TF-IDF feature extraction combined with linear classification architectures for 5-class movie sentiment analysis. Linear SVM established superior class-balanced generalization (champion by Macro F1: 50.27%), while Logistic Regression achieved peak test accuracy (63.96%), and Naive Bayes provided an ultra-efficient baseline (60.06%). Future extensions include exploring contextual word embeddings (such as Word2Vec or GloVe) and fine-tuning transformer-based architectures (e.g. DistilBERT) to better parse nuanced cinematic reviews.